

Trypanosoma cruzi

= Causative agent of Chagas disease

= Discovered by Carlos Chagas

= Named after mentor, Oswaldo Cruz

Chagas disease

=16-18 million infected

=100 million at risk

= 50,000 deaths annually

= leading cause of cardiac disease in South and Central America

Vectors of Chagas disease

Common Names

- triatomine bugs
- reduviid bugs
- assassin bugs
- kissing bugs
- cone nose bugs



Genera

- *Triatoma*
- *Rhodnius*
- *Panstrongylus*

Modes of transmission

Source	Comments
Vector	Triatomine bugs (contaminated feces)
Transfusion	Urban areas
Congenital	Any stage of <i>T. cruzi</i> infection [Premature labor, abortion, neonatal death]
Accidental	Ingestion of food contaminated with metacyclic trypomastigotes; laboratory accidents

Chagas disease: disease stages

Acute Phase

- active infection
- 1-4 months duration
- most are asymptomatic (children most likely to be symptomatic)

Indeterminate Phase

- 10-30 years of latency
- relatively asymptomatic with no detectable parasitemia
- seropositive

Chronic Phase

- 10-30% of infected exhibit cardiomyopathy or megasyndromes

Chagas disease: disease stages

Acute Phase features

- 1-2 week incubation period
- local inflammation
 - Romaña's sign
 - Chagoma
- symptoms can include: fever, malaise, lymphadenopathy, hepatosplenomegaly, nausea, diarrhea
- acute, often fatal, myocarditis develops in a few individuals
- high parasitemia in myofibrils (amastigotes)

Chagas disease: disease stages

Chronic Phase

= Cardiomyopathy (cardiomegaly)



= Megaviscerae (esophagus, colon)



Chagas disease: diagnosis

Parasite detection

- direct examination
- stained blood smears
- inoculation into mice
- in vitro culture
- xenodiagnosis
- PCR

Serological tests

- hemagglutination
- immunofluorescence
- ELISA
- complement fixation

Chagas disease: treatment

Acute stage

= nifurtimox (8-16 mg/kg/day, 60-90 days)

= benznidazole (5-7 mg/kg/day, 30-120 days)

Chronic stage

= treat symptoms